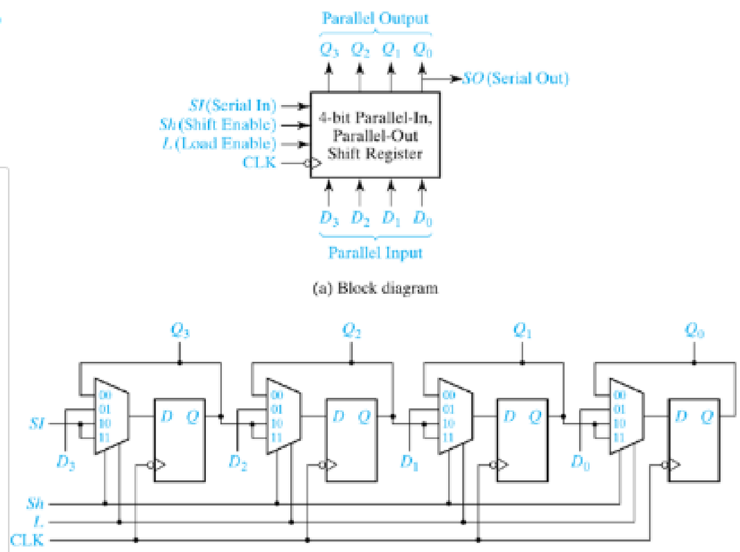
ECE 3441 Homework Unit 12 Name/ID#______________________

12.3, 12.6, 12.15, 12.36

12.3 Show how to modify the internal circuitry of the shift register of Figure 12-10 so that it will also shift to the left without external connections as in Problem 12.2. Replace Sh and L with A and B and let the register operate according to the following table:

Inputs		Next state				
A	B	Q_3^+	Q_2^+	Q_1^+	Q_0^+	Action
0	0	Q_3^+	Q_2^+	Q_1	Q_0^+	no change
0	1	SI	Q_3	Q_2	Q_1	right shift
1	0	Q_2	Q_1	Q_0	S_1	left Shift
1	1	D_3	D_2	D_1	D_0	load

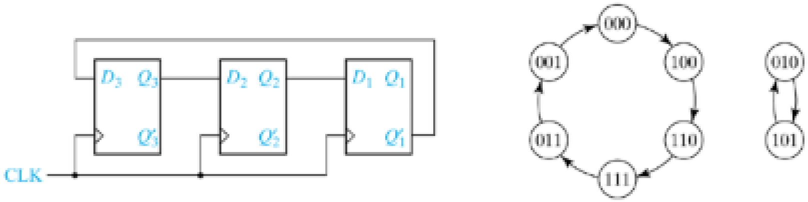


(a) Block diagram

(b) Implementation using flip-flops and MUXes

12.6 Design a circuit using D flip-flops that will generate the sequence 0 , 0 , 1 , 0 , 1 , 1 and repeat. Do this by designing a counter for any sequence of states such that the first flip-flop takes on this sequence. There are many correct answers, but do not duplicate states, because each state can have only one next state.

12.15 Construct a **4-bit** Johnson counter using **J-K** flip-flops. (See Figure 12-12 for a Johnson counter.) What sequence of states does the counter go through if it is started in state **0000** ? State **0110** ?



(a) Flip-flop connections © Cengage Learning 2014 (b) Transition graph

12.36 A M-F flip-flop behaves as follows:

If $MF = 01$, the flip-flop changes state.

If $MF = 11$, the flip-flop is set to $Q = 0$.

If $MF = 00$, the flip-flop is set to $Q = 1$.

The input combination $MF = 10$ is not allowed.

a. Give the characteristic (next-state) equation for this flip-flop.

b. Complete the table, using don't-cares where possible.

Q	Q^+	M	F
0	0		
0	1		
1	0		
1	1		

c. Realize the following next-state equation for Q using a MF flip-flop:
$Q^+ = CQ + DQ'$. Find equations for M and F .